

VivekSingh

+91-8929554634 | singhviv2712@gmail.com | LinkedIn: <https://www.linkedin.com/in/vivek-singh2712>

PROFESSIONAL SUMMARY

Strategic AI & Automation Engineer with 4 years of experience architecting intelligent, high-scale applications. Expert in building robust back-end frameworks using **Python, Django, and FastAPI**, seamlessly integrated with **ReactJS** Front-End and **UI/UX** principles. Specialized in **GenAI** and **Agentic AI**, leveraging **Transformers Architecture, U-Net architectures**, and **Deep Learning** to deliver **LLM** solutions through **Fine-Tuning** and **RAG**. Proven track record in deploying **NLP** models and **AI algorithms** using **TensorFlow** and **Keras**, supported by **Vector DB, Oracle**, and **Docker** for scalable enterprise automation and high-performance, **Full-Stack** architectures.

EXPERIENCE

Automation Engineer | Bharti Airtel

Mar 2022-Present

- **Generative AI & LLM Operations (MOP.ai):** Engineered a specialized **Transformer-based** architecture to automate the generation of Methods of Procedure (MOP). Developed a decoder-based inference system integrated with **RLHF-inspired adaptive learning** and reasoning modules, enabling the system to evolve dynamically based on user feedback. Developed LLM Model using Transformers, TensorFlow, Keras.
- **Agentic AI & Unified Analytics:** Developed the **Unified Data Inventory**, a cross-domain ticketing platform utilizing **Agentic AI** via **Model Context Protocol (MCP)** servers. Integrated **Natural Language Search** and specialized AI agents to perform deep-dive trend reporting and automated incident analysis across thousands of Remedy tickets.
- **Predictive Operations (Nexus):** Architected with a high-performance dashboard using **FastAPI** and **ReactJS**, featuring an integrated **LSTM (Long Short-Term Memory)** inference engine. This system provides real-time predictive analytics for ticket aging and SLA breach volumes, enabling proactive resource allocation.
- **Computer Vision & Generative Topology (Pixel Engine):** Repurposed **U-Net image segmentation** architectures for generative network design. Adapted the model to identify and generate network components (routers, switches, links) on a digital canvas, evolving the U-Net into a decoder for text-to-image network topology generation.
- **Autonomous Network Optimization:** Developing a **Deep Learning** framework to optimize network performance through real-time traffic analysis. The system utilizes predictive modeling to execute dynamic routing adjustments, significantly reducing latency and preventing network outages.
- **Vendor-Agnostic Automation (Airtel Image Manager):** Designed and deployed custom interfaces to replace Cisco SWIM and MOPA. These solutions automated firmware upgrades and script deployments across a multi-vendor environment (**Cisco, Nokia, Huawei, Juniper, Ciena**), drastically reducing feature-request lead times.
- **Enterprise Integration & Security:** Orchestrated the integration of **Netcool** and **Remedy** for automated fault-to-ticket workflows. Strengthened enterprise security by implementing **SSO with LDAP Authentication**, providing centralized access management across the internal application ecosystem.
- **Real-Time Data Streaming (Kafka):** Integrated **Apache Kafka** to facilitate high-throughput, low-latency data pipelines. This architecture supports real-time processing of network telemetry, enhancing the

scalability and resilience of Airtel's data ecosystem.

- **Data Lake Engineering:** Developed a centralized **Data Lake** solution for Call Detail Records (CDR) and Syslog data. Engineered the pipeline to capture and analyze MSAN, DSL, and BNG interface metrics and Syslog data providing a web-based visualization layer for critical troubleshooting and decision-making.

PROJECTS

ORBITAI

Project Overview: Designed and developed an intelligent Inventory SOP (Standard Operating Procedure) platform focused on operational knowledge discovery, document intelligence, and network audit automation. The platform centralizes SOP documents, operational procedures, historical records, and audit artifacts into a searchable knowledge-driven system. A knowledge graph layer models relationships between devices, services, incidents, SOP steps, dependencies, and operational entities. This enables engineers to perform contextual search instead of relying only on keyword matching. The system combines graph-based deep search, semantic retrieval, recommendation intelligence, and LLM-powered agent workflows. It helps users quickly locate relevant SOPs, understand dependencies, discover related operational documents, and generate actionable insights. In addition, an Audit Toolkit was developed to execute issue-specific audit workflows against network devices, collect operational logs, analyze deviations, estimate impact, and produce automated LLM-based analysis for faster troubleshooting and RCA.

- **Impact:** Reduced script modification and deployment time from up to six months to a matter of hours, significantly lowering costs and operational delays.

▮ **Technical Workflow:** The audit engine collects live operational logs from network devices and compares runtime behavior against expected SOP-defined states. It identifies deviations, anomalies, and configuration mismatches that may affect service stability. Based on dependency relationships in the knowledge graph, the system performs impact analysis to determine affected services, devices, and operational domains. Using Llama 3.2 and LangChain, the platform generates contextual analysis of collected logs, highlighting probable root-cause indicators, operational observations, and recommended remediation actions. This enables faster troubleshooting, audit validation, and decision-making for network operations teams.

▮ **Tools & Technologies:** Python, ReactJS, FASTAPI, Oracle, REST APIs, LangChain, LLAMA3.2: latest, A-Star Search Algo., Heap,

NetVision: Data Lake & Syslog GENAI/Predictive AI

- **Project Overview:** Implemented a robust Data Lake and interface solution for managing syslog data containing network alarms, operational events, and alarm categorization across network elements. The platform was designed to centralize operational telemetry from routers and switches, enabling real-time monitoring, pattern detection, and intelligent operational analysis.
- **Impact:** Centralized previously siloed network data into a unified analytical platform, providing a scalable foundation for troubleshooting dropped connections, identifying abnormal operational behavior, and enabling data-driven decision-making for network performance and service stability.
- **Technical Workflow:** The system captures and stores massive volumes of syslog data, including events, alarms, source information, and alarm categories, into a centralized Data Lake. It features a web-based visualization layer and a suite of APIs for data retrieval, enabling real-time monitoring of BRAS/BNG and QFX switch performance. Historical and streaming operational data are continuously processed for correlation analysis, anomaly detection, and root cause interpretation.
- **Pattern Identification:**
A dedicated pattern identification layer was implemented to detect relationships, operational anomalies, and fault patterns from live syslog streams.
- **Correlation Analysis:** Correlation scoring is derived using weighted operational parameters — $\text{Event} \times 0.35$, $\text{Alarms} \times 45$, $\text{Source} \times 20$ — to identify related events, recurring alarm clusters, and dependency-linked operational patterns.
- **Heatmap Visualization:** Correlation outputs are represented through heatmap-based visualization to highlight concentration zones, recurring fault hotspots, and high-impact operational areas.
- **Anomaly Detection:** Implemented DRAIN-based anomaly detection for syslog template extraction, abnormal behavior identification, and event deviation detection.
- **RCA (Root Cause Analysis):** Correlated events, anomaly signatures, and operational dependencies are analyzed to identify probable root-cause indicators, impacted network domains, and recommended remediation actions.
- **Tools & Technologies:** OracleDB, FASTAPI, LSTM(RNN), ReactJS, Deep Learning, LLM, Agentic AI, MCP Server, Drain Algorithm, Deep Agents

Airtel Image Manager (AIM)

- **Project Overview:** Designed and implemented a custom interface to replace the Cisco SWIM interface, enabling efficient and automated

upgrades across multiple vendor routers, including Cisco, Nokia, Huawei, Juniper, and Ciena.

- **Impact:** Reduced script modification and deployment time from up to six months to a matter of hours, significantly lowering costs and operational delays.
- **Technical Workflow:** The interface operates in two phases. In the first phase, the required IP addresses and change request numbers be uploaded, after which a single button click initiates, the iOS copying process. This process fetches data from a MySQL database and executes REST API scripts on target routers, logging results in real-time. In the second phase, routers with successfully copied iOS are queued for upgrades. This upgrade process is fully automated and restricted to nighttime hours (12 AM - 5 AM) for minimal disruption, with the interface auto-enabling for this period. SSO integration ensures secure, single-sign-on access.
- **Tools & Technologies:** Python Scripting, ReactJS, Django (DRF), MySQL, REST APIs

AI Insights: -

- **Project Overview:** Designed and implemented a custom AI-powered chatbot specifically for network engineers. The system provides intelligent, context-aware troubleshooting and optimization recommendations by combining domain-specific reasoning with real-time data retrieval.
- **Impact:** Empowered operations teams with instant access to complex network insights, reducing troubleshooting time and improving incident response through an adaptive, self-learning AI loop.
- **Technical Workflow:** The system utilizes Retrieval-Augmented Generation (RAG) to query a ChromaDB vector database for relevant network documentation and logs. This data is fed into the AirLM model for reasoning. A reinforcement learning loop captures user feedback to fine-tune the model, continuously improving the relevance of responses.
- **Tools & Technologies:** AirLM (Custom LLM), RAG, ChromaDB (Vector DB), FastAPI, ReactJS, Reinforcement Learning.

AirLM: Domain-Specific Transformer LLM

- **Project Overview:** Developed a custom Large Language Model (LLM) from the ground up, tailored specifically for telecommunications and network engineering domains. Unlike generic models, AirLM is optimized for processing technical network language and complex reasoning tasks.
- **Impact:** Created a portable, high-performance inference system that provides deeper, more accurate domain-specific insights than off-the-shelf models, with full control over data privacy and architecture.
- **Technical Workflow:** Built using a stacked Transformer architecture (decoder-based), the pipeline includes custom tokenization, embeddings, and positional encoding. The model incorporates RLHF-inspired adaptive learning and reasoning modules. It is served via a FastAPI inference engine with exported config.json and vocab.json for reproducible deployment.
- **Tools & Technologies:** TensorFlow, Keras, Python, FastAPI, ReactJS, Transformer Architecture, RLHF.

PixelEngine: Generative Network Topology & Segmentation

- **Project Overview:** Repurposed a trained U-Net image segmentation model to function as a generative engine for network design. The project explores the intersection of computer vision and network architecture to visualize complex infrastructures.
- **Impact:** Automated the generation of network topology diagrams, converting text-based prompts into visual representations of routers, switches, and links, facilitating faster planning and documentation.
- **Technical Workflow:** The U-Net's segmentation capabilities were adapted to identify distinct network components as "segments" on a canvas. For generative tasks, the U-Net acts as a decoder within a larger GAN or Diffusion framework, where latent vectors from text prompts guide the model to construct high-fidelity network topology images.
- **Tools & Technologies:** U-Net Architecture, GANs/Diffusion Models, Deep Learning, Image Segmentation, Python.

MOP.ai: Automated Method of Procedure Engine

- **Project Overview:** Built an AI-driven application to automate the creation of complex Method of Procedure (MOP) documents. The tool replaces manual drafting with an intelligent engine that understands standardized network change protocols.
- **Impact:** Dramatically reduced the administrative overhead for engineers, ensuring 100% consistency in MOP documentation and reducing the risk of human error during network changes.
- **Technical Workflow:** Established a comprehensive repository of standardized MOPs to serve as a training baseline. A Transformer-based architecture was trained on this repository to learn the structural and technical nuances of MOPs. The model was then integrated into a production pipeline to generate documents from minimal input variables.
- **Tools & Technologies:** Python, Document Automation, Natural Language Processing. ReactJS, FASTAPI

Unified Data Inventory: Agentic AI Analytics Platform

- **Project Overview:** Engineered a centralized ticketing analytics platform that aggregates data across four domains: Homes, Infra, Service, and Passive. It features an advanced AI layer for deep-dive reporting and natural language data retrieval.

- **Impact:** Provided stakeholders with a "single source of truth" for operational health, enabling instant data retrieval through AI search and automated trend analysis that previously took hours to compile.
- **Technical Workflow:** The platform uses Model Context Protocol (MCP) servers to power specialized AI Agents. These agents perform cross-domain trend analysis and generate insights. Users can interact with the data via Natural Language Search, which translates queries into granular ticket data retrievals in real-time.
- **Tools & Technologies:** Agentic AI, MCP Servers, AI Search (Semantic Search), ReactJS, Python, ML Prediction, FASTAPI.

NOC Beats: Data Lake & CDR Analysis

- **Project Overview:** Implemented a robust Data Lake and interface solution for managing Call Detail Records (CDR) and monitoring MSAN/BNG network elements.
- **Impact:** Centralized previously siloed network data, providing a scalable foundation for troubleshooting dropped connections and making data-driven decisions regarding network performance.
- **Technical Workflow:** The system captures and stores massive volumes of CDR data, including DSL IDs and interface metrics, into a centralized Data Lake. It features a web-based visualization layer and a suite of APIs for data retrieval, allowing for real-time monitoring of BNG and MSAN performance.
- **Tools & Technologies:** Oracle DB, Restful API, Django, ReactJS.

Automated Remedy Ticketing for Network Events

- **Project Overview:** Developed an automated system to detect network issues and directly create tickets in Remedy for incident tracking, bypassing the need for Netcool. This solution continuously monitors network performance, and upon detecting any anomalies, it directly raises a ticket in Remedy, ensuring critical issues are promptly addressed.
- **Impact:** Enhanced operational efficiency and reduced manual workload by automating ticket creation for network incidents. This system ensures that all detected network issues are documented and tracked in Remedy without human intervention, allowing faster response times and reducing the risk of missed incidents.
- **Technical Workflow:** The system periodically runs a monitoring script using a cron job to scan network anomalies. When an issue is detected, the system automatically generates a Remedy ticket through direct API integration. This setup eliminates the need for intermediary tools, streamlining the incident management process from detection to ticketing.
- **Tools & Technologies:** ReactJS, Django, Python Scripting, Remedy API, REST API.

SKILLS

Python, Django, FASTAPI, TensorFlow, RESTFUL API, Keras, Vector DB, Oracle DB, Docker, MySQL, Transformers, DSA
Transformer’s Architecture, U-NET Architecture, Deep Learning, Machine Learning, Agentic AI, GENAI, ReactJS, UI/UX, A2A, MCP Server, n8n Workflow, NLP, SSO, LDAP, GCP, Jenkins, GIT, Deep Agents, Ai Agents

EDUCATION

B.tech – Electrical and Electronics Engineering

Aug 2018 - Aug 2022

Galgotia’s College of Engineering and Technology
CGPA- 7.22